

AI / Machine Learning Engineer – Technical Skills Assessment

Computer Vision Focus

Overview

This assessment evaluates **AI / Machine Learning Engineers (3+ years)** with a strong focus on **computer vision**, end-to-end ML workflows, and production-oriented thinking.

This is **not**:

- A data analyst task
- A research exercise
- A prompt-engineering role

This **is** a practical ML engineering task covering:

- Data understanding and preprocessing
- Model selection and training
- Evaluation and error analysis
- Deployment-ready inference logic

You are **not expected** to build a production-scale system. A clean, well-reasoned prototype is sufficient.

Time Expectations

- **Expected effort:** ~48 hours
- **Hard cap:** 2 days
- Follow-up technical discussion: **~45 minutes**

Submission Instructions

- Push your code to a **GitHub repository**
- Share the **Git repository URL**
- Include a **README** explaining your approach, decisions, and results

Optional but good to have:

- A Dockerized inference service
- A short demo video or walkthrough

Assessment Objective

Build a **computer vision model** for a real-world problem and demonstrate **ML engineering discipline under realistic constraints**.

The emphasis is on:

- Correct problem framing
- Sensible model choices
- Clean evaluation practices
- Production-aware inference design

Core Task: Computer Vision Problem

Problem Statement

Choose **one** of the following **computer vision tasks** using a **public dataset**:

1. **Image classification**
2. **Object detection**
3. **Image segmentation** (optional, higher difficulty)

The dataset should be **non-trivial** (avoid MNIST or CIFAR-10).
Clearly explain **why you chose this dataset** and how it fits the task.

Required Deliverables

1. Data Understanding & Preparation

Requirements:

- Inspect and describe:
 - Dataset size
 - Label format
 - Class distribution
- Apply appropriate:
 - Image preprocessing
 - Data augmentation (if applicable)
- Identify and handle:
 - Class imbalance
 - Noisy or inconsistent labels (if observed)

Evaluation Focus:

- Practical understanding of CV data
- Sensible preprocessing choices
- Awareness of dataset limitations

2. Model Selection & Training

Requirements:

- Train **at least two models**, for example:
 - A simple baseline
 - A stronger model (e.g., fine-tuned pre-trained network)
- Use transfer learning **only when justified**
- Perform basic hyperparameter tuning

Constraints:

- Model choice must be **defensible**
- Training strategy should reflect **realistic compute limits**

Compute & Resource Constraints

- This task is designed to be completed **without access to high-end GPUs**
- Solutions that run on **CPU, laptop GPUs, or free-tier cloud resources** are fully acceptable
- Candidates will **not** be evaluated on raw accuracy, training time, or model size
- Model and training decisions should explicitly reflect **compute awareness**

We care about **engineering judgment and reasoning**, not brute-force compute.

3. Evaluation & Error Analysis

Requirements:

- Proper train/validation/test split
- Use appropriate evaluation metrics for the selected task
- Perform **qualitative and quantitative error analysis**, such as:
 - Visualizing misclassified or poorly predicted samples
 - Identifying common failure modes

Evaluation Focus:

- Correct evaluation methodology
- Insightful, data-driven analysis
- Clear improvement hypotheses

4. Inference & Deployment (Mandatory – Live Deployment Required)

This project must be deployed to a publicly accessible environment.

Requirements:

- The trained model must be exposed through a live, publicly accessible endpoint.
- A deployment link must be provided in the README.
- The endpoint must allow inference via:
 - REST API (preferred), or
 - Simple web interface for image upload and prediction.
- The deployed service must be functional at the time of submission.

Acceptable platforms include (but are not limited to):

- Render
- Railway
- Hugging Face Spaces
- AWS / GCP / Azure (free tier is acceptable)
- Any equivalent hosting platform

Minimum expectation:

A working, publicly accessible deployment link that performs inference using the trained model.

Submissions without a live deployment link will be considered incomplete.

5. Engineering Quality

Requirements:

- Project structure beyond a single notebook
- Clear separation of:
 - Data processing
 - Training
 - Inference
- Configurable parameters (paths, model selection, etc.)
- Basic logging and error handling

Evaluation Focus:

- ML engineering maturity
- Code readability and maintainability
- Ability to move beyond experimentation

Technical Skills Being Assessed

- **Computer Vision:** preprocessing, augmentation, evaluation
- **Deep Learning:** CNNs, transfer learning, fine-tuning
- **ML Fundamentals:** metrics, validation, error analysis
- **Engineering:** clean code, structure, reproducibility
- **Deployment Thinking:** inference pipelines, APIs, Docker familiarity
- **Problem Solving:** data-driven iteration and tradeoffs

Evaluation Criteria

Will not be evaluated:

- UI or frontend work
- Cloud-scale infrastructure
- Training extremely large models
- Chasing state-of-the-art benchmarks

Will be evaluated:

- Correct ML methodology
- Quality of reasoning and tradeoffs
- Practical computer vision understanding
- Clean, maintainable ML code
- Deployment readiness
- Ability to clearly explain failures and decisions

A simpler model with **strong analysis and clean engineering** will score higher than a complex model trained blindly.

Final Note

This task is designed to reflect **how you work as an ML engineer under real-world constraints**.

If additional compute were available, briefly describe:

- How you would scale training
- What model or data improvements you would explore

We care more about **thinking, correctness, and discipline** than squeezing out marginal accuracy gains.